

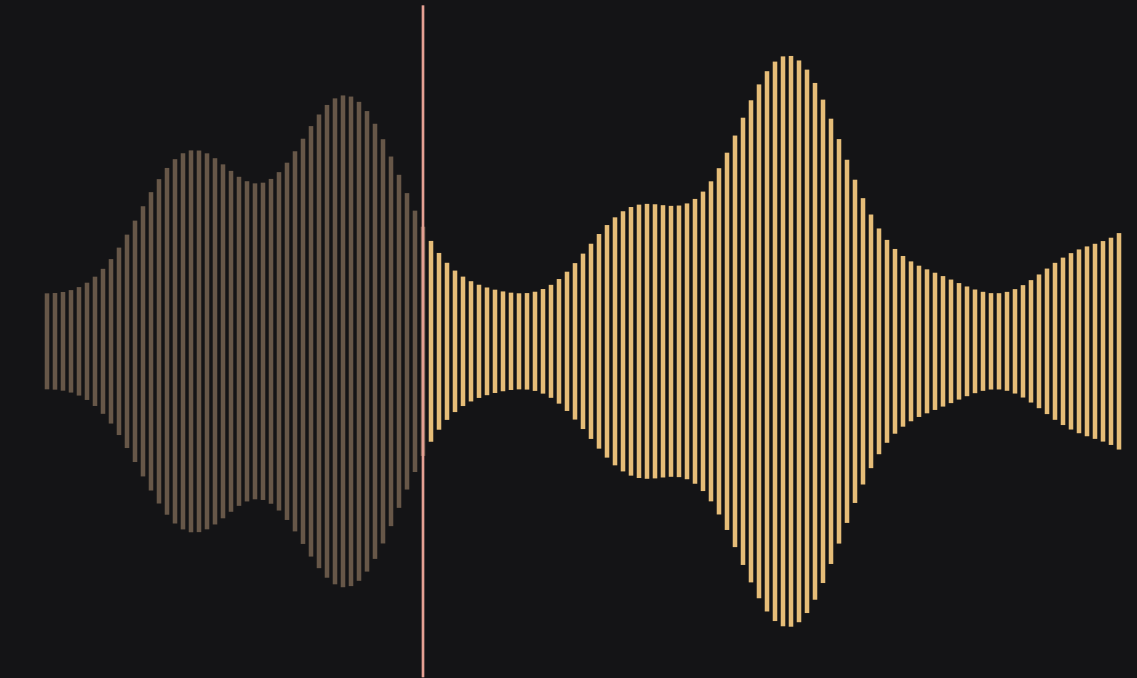
SonicCraft

Audio editing through
Signals & Systems

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Python desktop application



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Why SonicCraft?

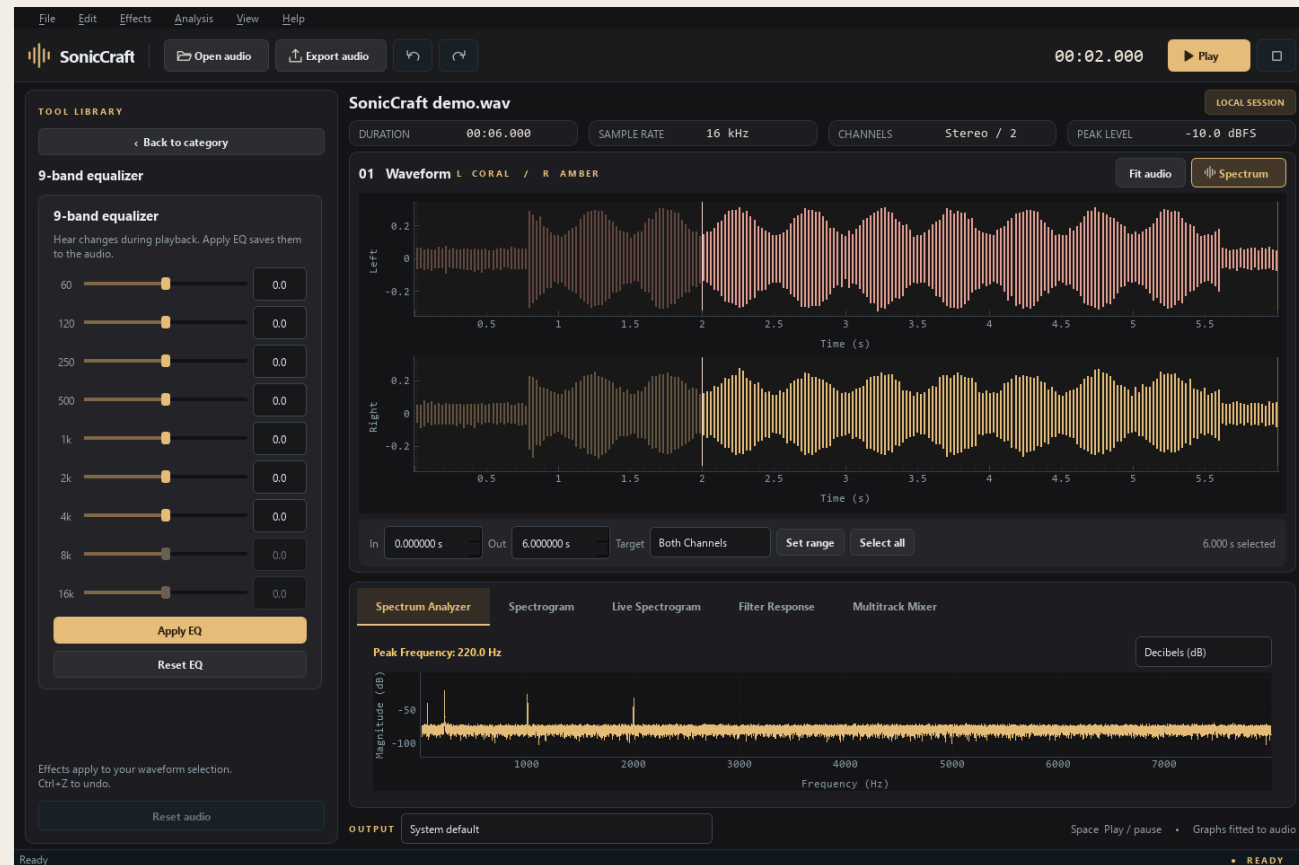
Make a signal operation visible and audible.

THE PROBLEM

A noisy recording
hides useful sound.

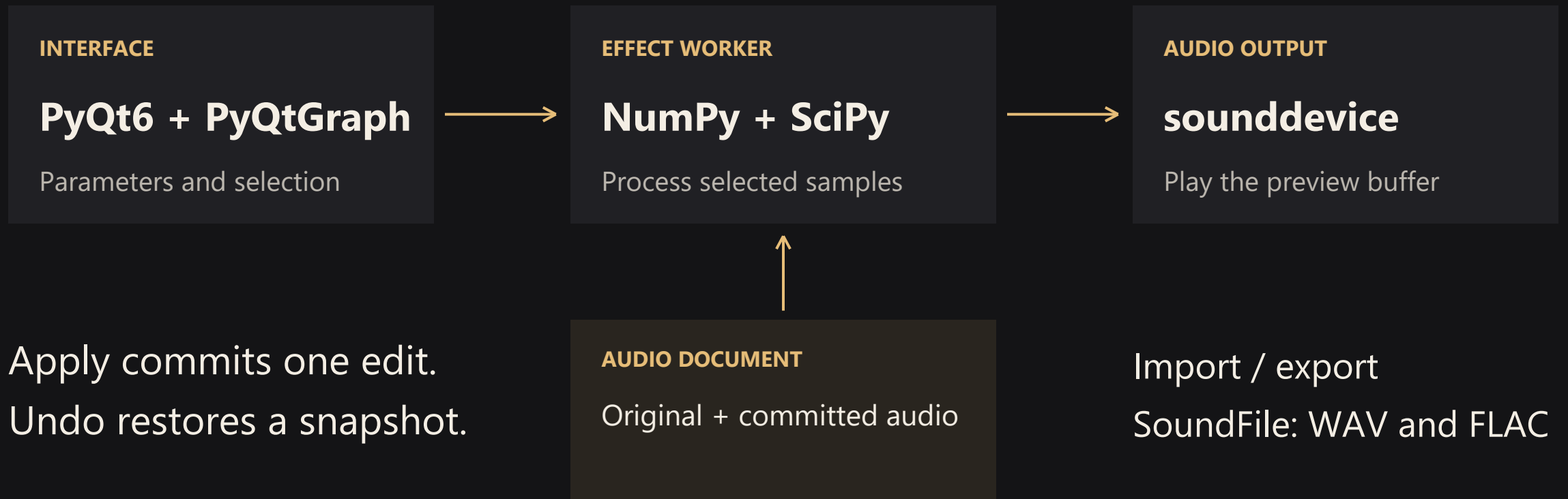
OUR APPROACH

Inspect the signal.
Adjust a parameter.
Hear the result.



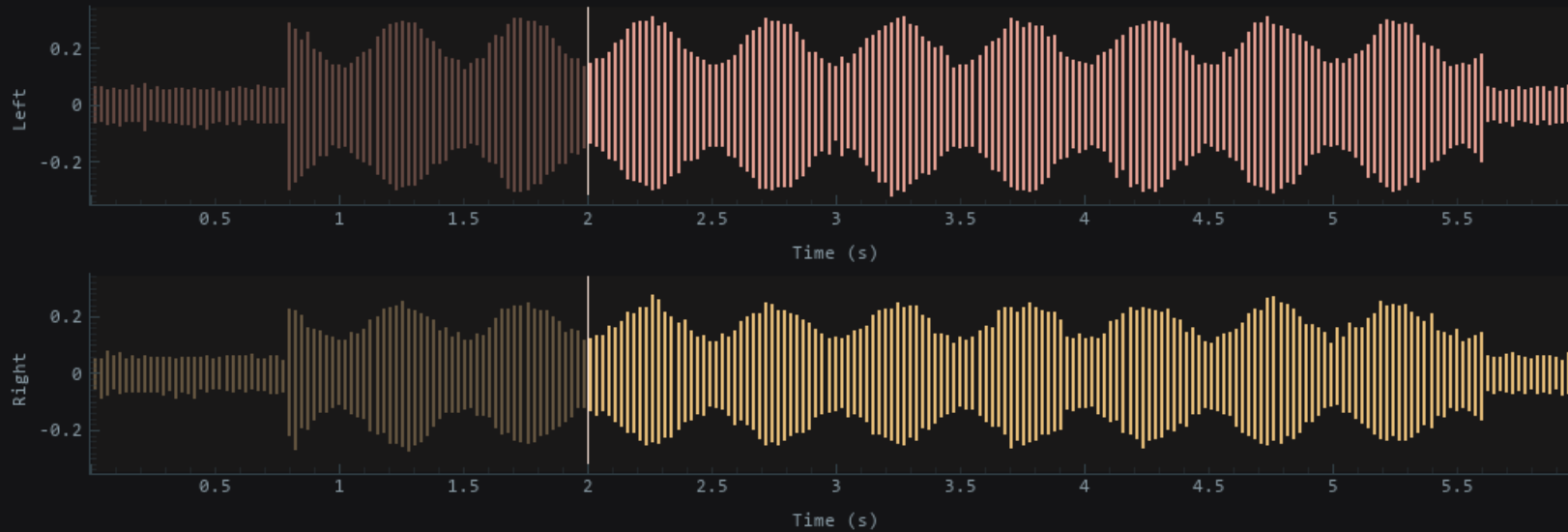
How the application works

One audio document connects the controls, DSP and playback.



Waveform and selection

Choose which part of the audio to edit.



THEORY CONNECTION

$$n = \text{round}(tF_s)$$

Time corresponds to sample position.

Gain

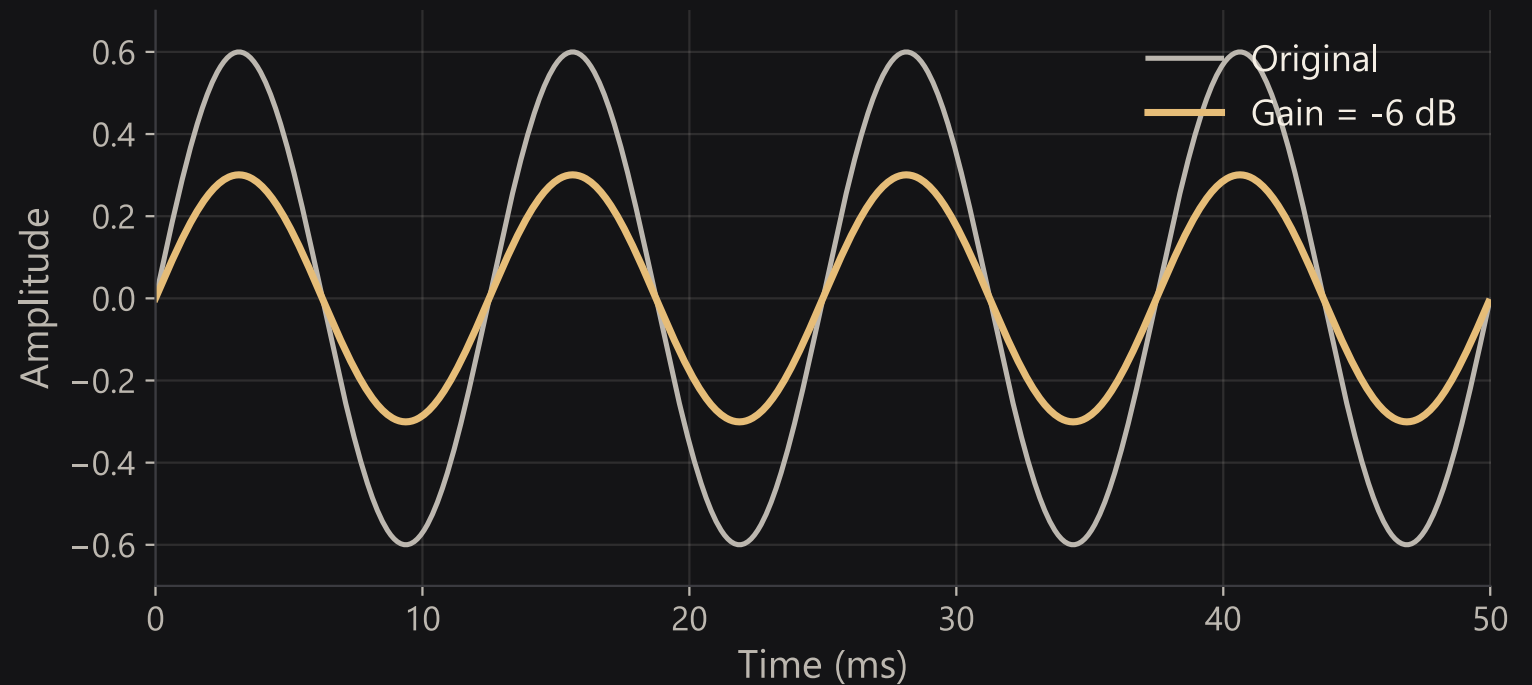
Make audio louder or quieter.

THEORY CONNECTION

Amplitude scaling

Multiply each sample by a constant.

$$y[n] = 10^{G/20} x[n]$$



Frequency filters

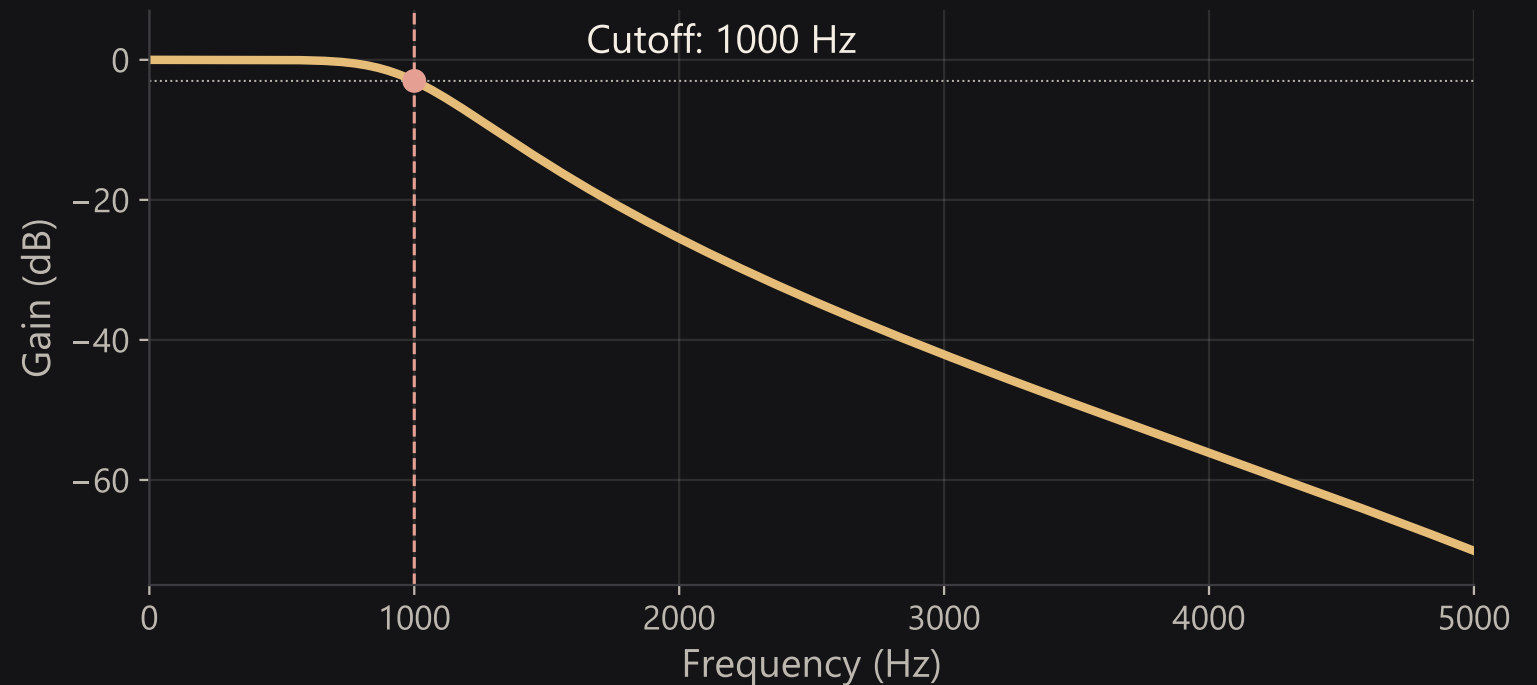
Keep some frequencies and reduce others.

THEORY CONNECTION

LTI systems

The frequency response shapes the sound.

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$



9-band equalizer

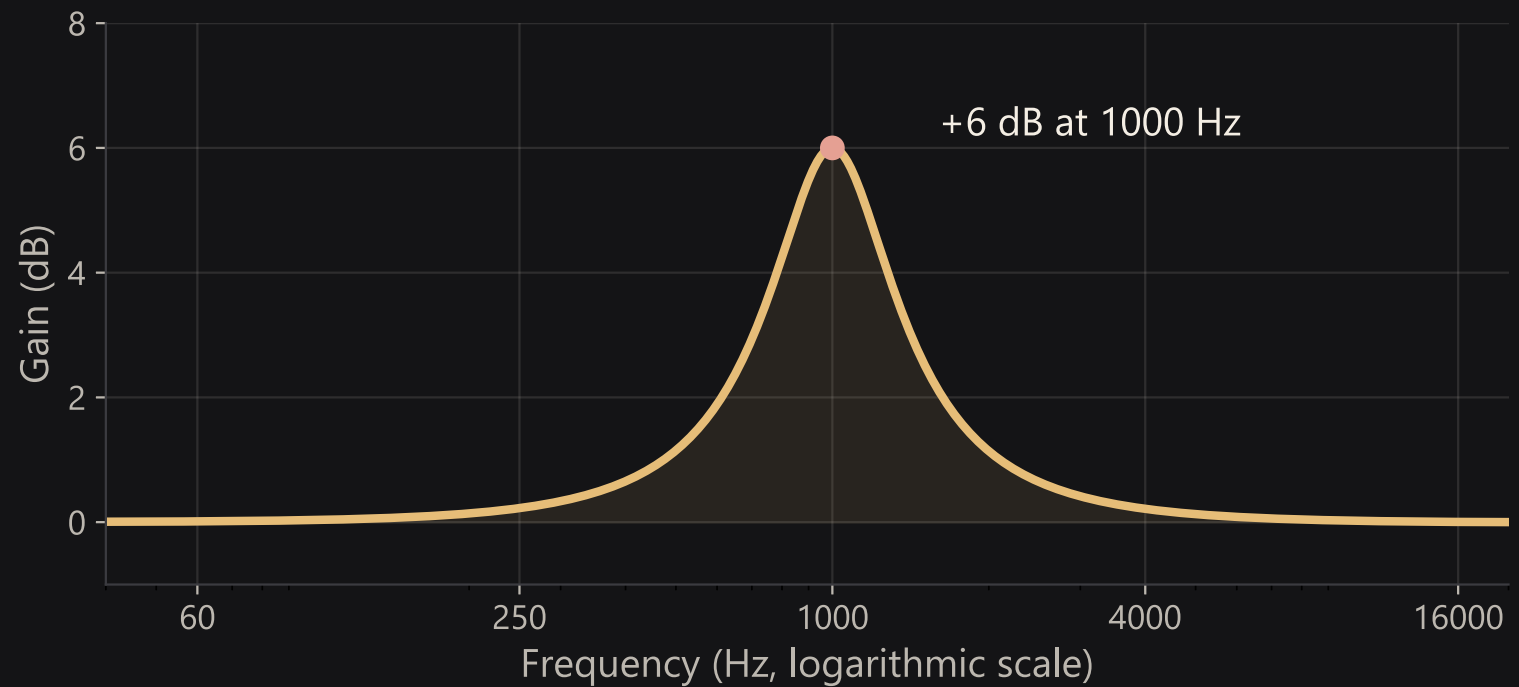
Adjust bass, midrange and treble.

THEORY CONNECTION

Cascaded filters

Each band controls part of the spectrum.

$$H(z) = \prod_{b=1}^B H_b(z)$$



FFT spectrum

See which frequencies are present.

THEORY CONNECTION

Fourier decomposition

Break a sound into frequency components.

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}$$



Spectrogram

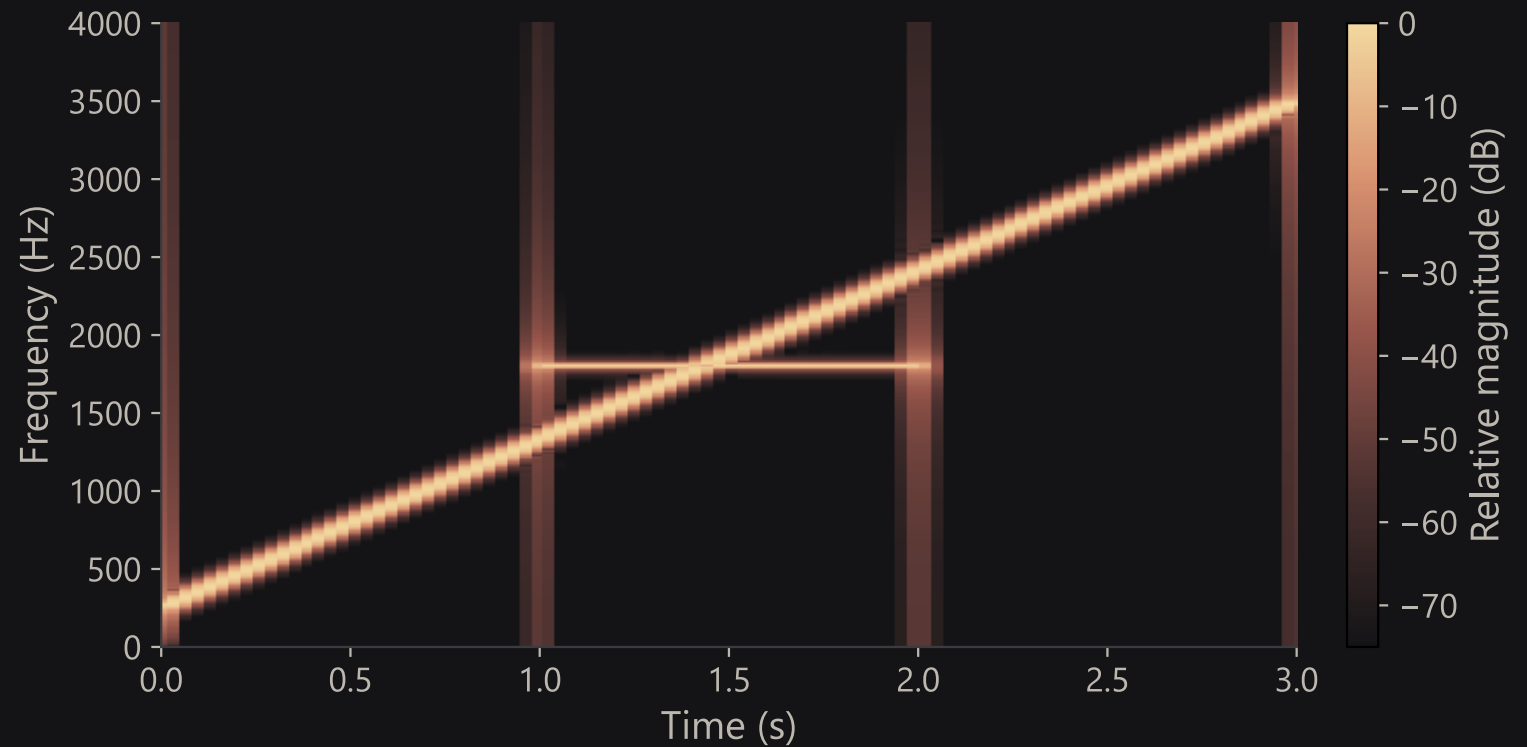
See how frequencies change over time.

THEORY CONNECTION

Windowed Fourier analysis

Analyze short sections of the sound.

$$\Delta f = \frac{F_s}{N}, \quad \Delta t = \frac{R}{F_s}$$



Noise reduction

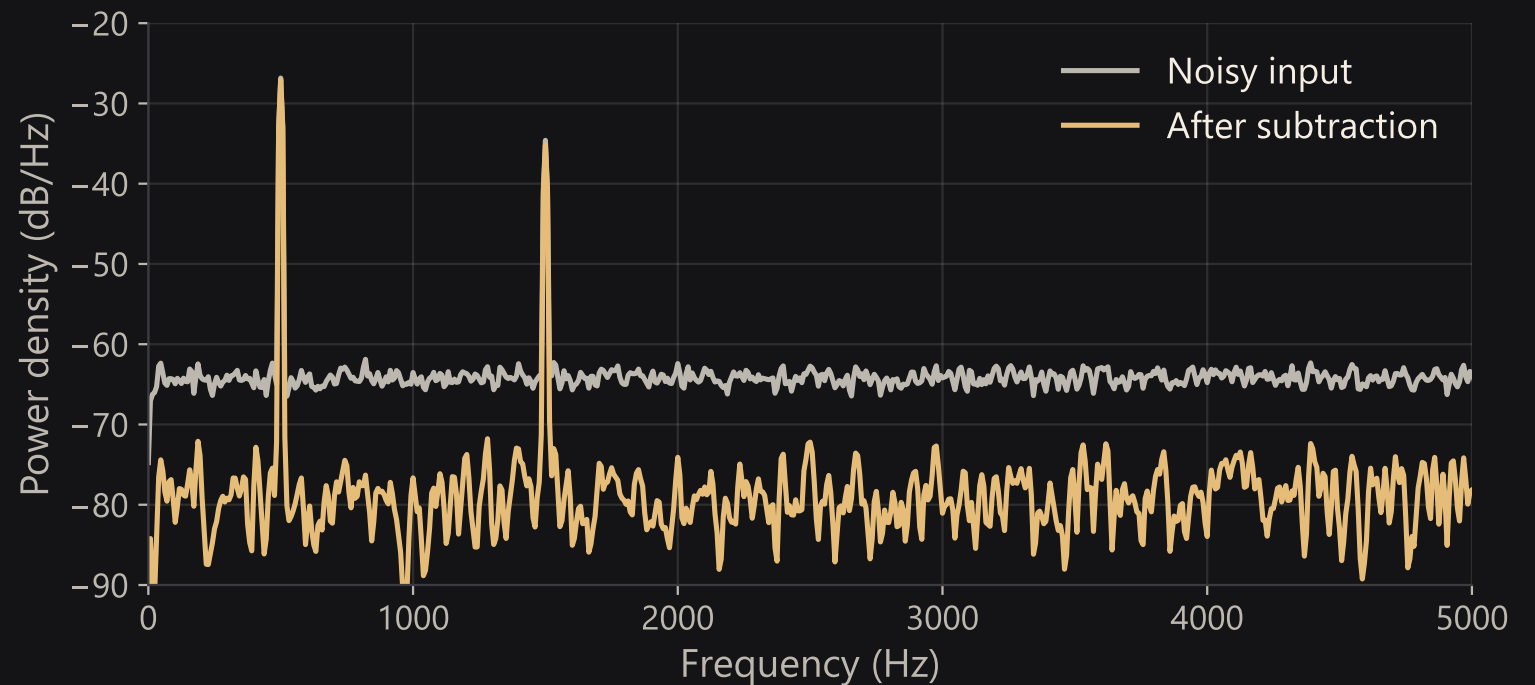
Reduce unwanted background sound.

THEORY CONNECTION

Spectral subtraction

Estimate the noise, then reduce its spectrum.

$$\hat{A} = \max(A - \alpha D, \beta A)$$



Preview, commit and recover

Parameter changes are temporary until Apply saves them to the audio.

DESIGN CHOICES

Responsive previews

Worker renders; callback keeps playing.

Controlled transitions

10 ms crossfade between audio buffers.

Two reset scopes

Tool reset clears settings. Main reset restores audio.

77

automated tests pass

NUMERICAL CHECKS

+6.000 dB

measured at the EQ center frequency

Preview delay includes rendering time.

Live demonstration

One six-second clip. A visible and audible result.

00-10 s	Open + play	Hear the stereo signal and background noise.
10-25 s	Change EQ	Move 1000 Hz, hear the preview, then Apply EQ.
25-40 s	Compare resets	Reset EQ keeps the edit. Reset audio restores the source.
40-65 s	Reduce noise	Capture 0-0.7 s. Select all. Preview subtraction and Apply.
65-80 s	Inspect + export	Generate the spectrogram, then export the processed WAV.

Questions

SonicCraft

A desktop application built around samples, filters and Fourier analysis.

github.com/mehrab4226/SonicCraft

Sample editing

Work directly with sample indices and selected intervals.

Trim

Keep only samples a through $b - 1$.

$$y[n] = x[n + a], \quad 0 \leq n < b - a$$

Reverse

Reverse time within the selected interval.

$$y[a + m] = x[b - 1 - m]$$

Silence

Zero the interval without changing its duration.

$$y[n] = 0, \quad a \leq n < b$$

Delete

Remove the interval and join the two sides.

$$y = x[:a] \parallel x[b:]$$

Fades

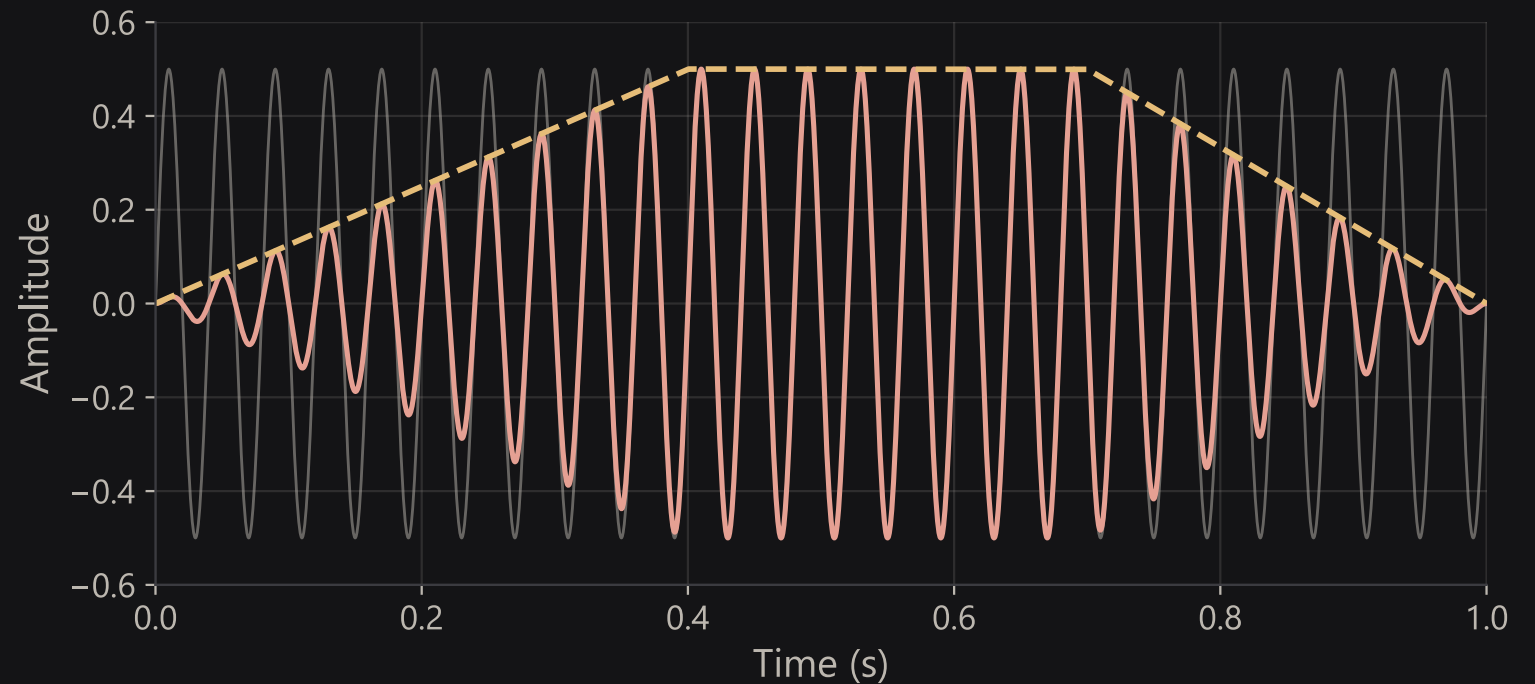
Make the start or end of audio gradual.

THEORY CONNECTION

Amplitude envelope

The amplitude changes over time.

$$y[n] = a[n]x[n]$$



Peak normalization

Set the selection peak to a chosen level.

THEORY CONNECTION

Amplitude scaling

The input peak determines the gain.

$$y[n] = \frac{A_{\text{target}}}{\max_m |x[m]|} x[n]$$

APP BEHAVIOR

-1 dBFS

Default peak target

Adjusts the largest sample.
Silence remains unchanged.

Multitrack mixing

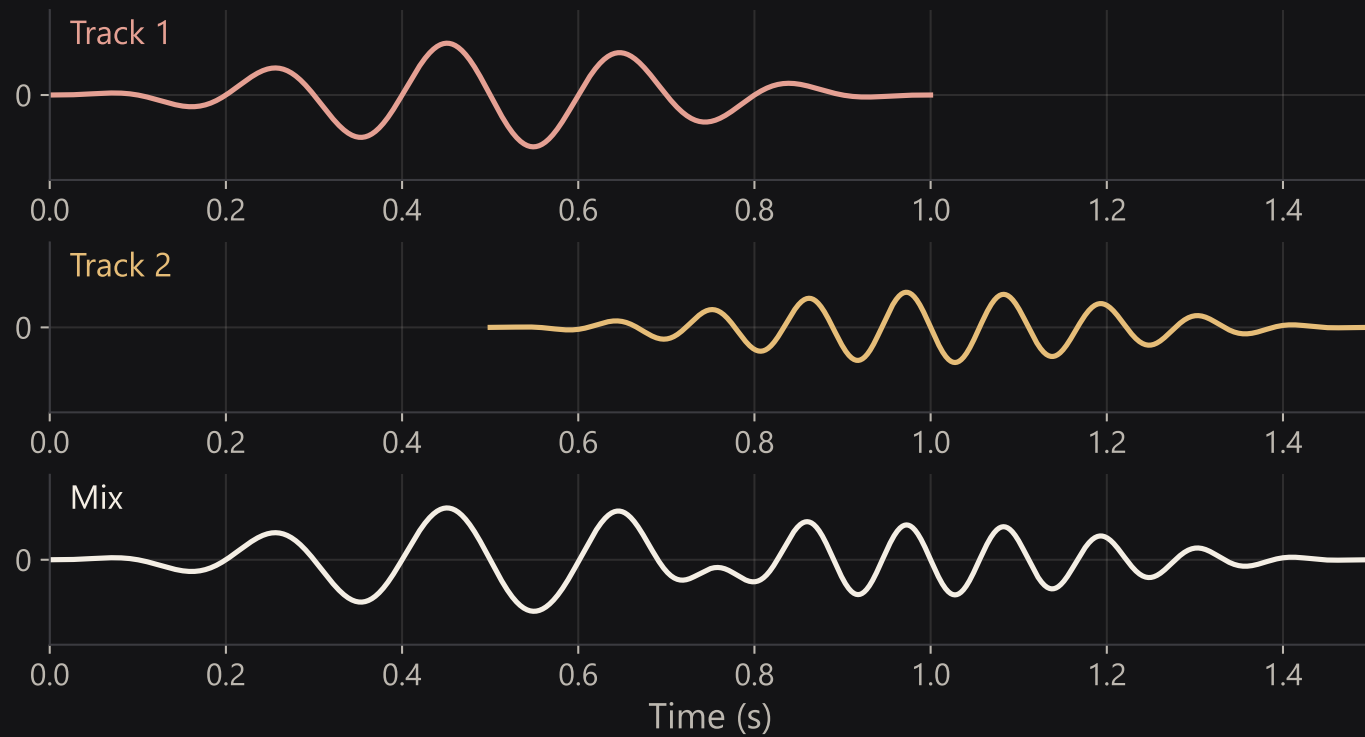
Combine several audio tracks.

THEORY CONNECTION

Superposition and time shift

Add signals with different
levels and offsets.

$$y[n] = \sum_i g_i x_i[n - d_i]$$



Spectral editing

Mute a time-frequency region.

THEORY CONNECTION

STFT masking

Remove selected components, then reconstruct.

$$\hat{X}[m, k] = M[m, k]X[m, k]$$

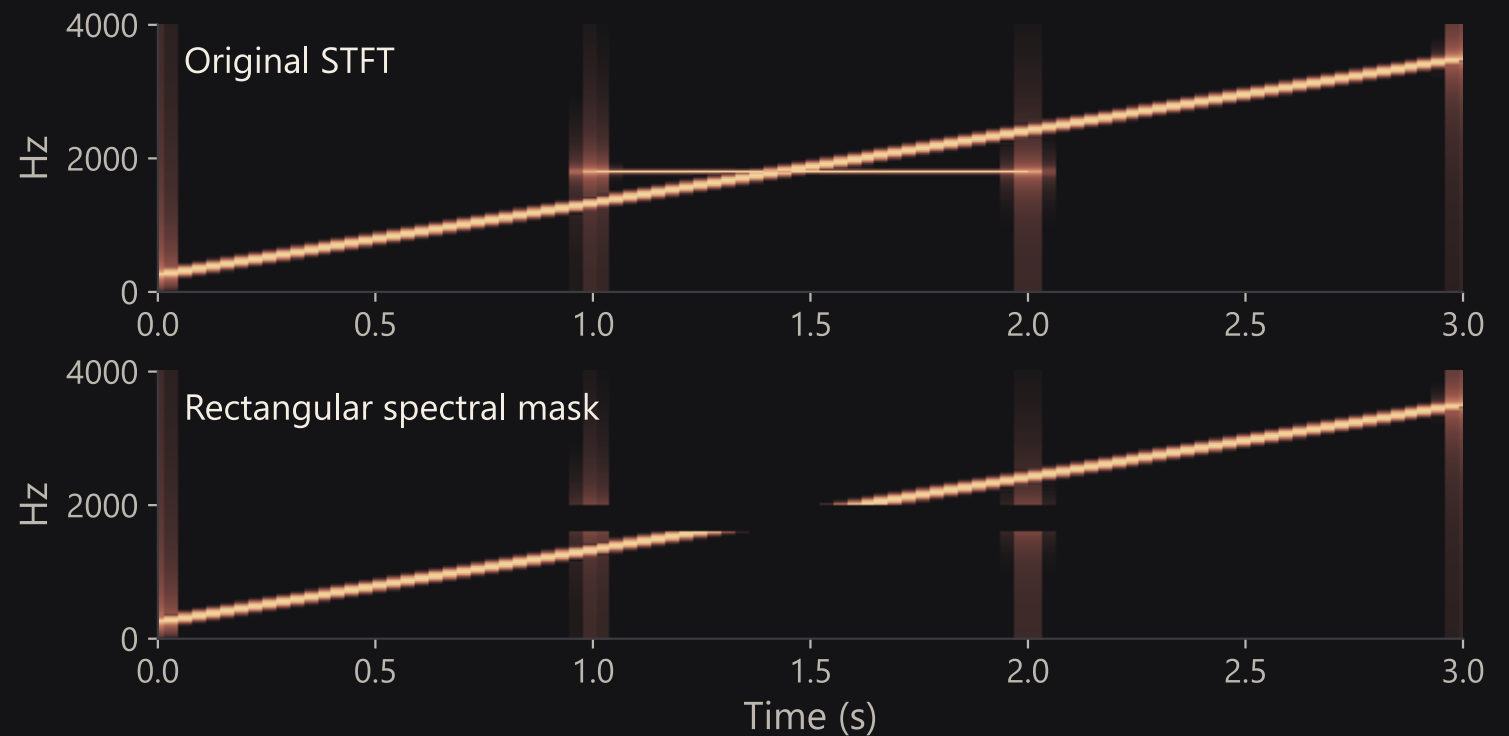
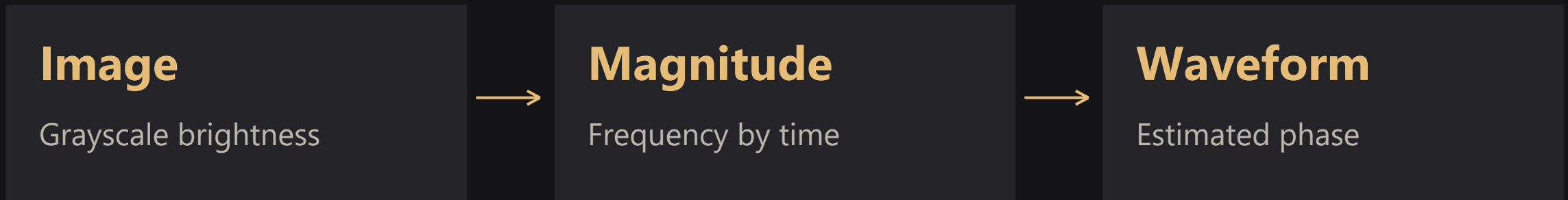


Image-to-audio synthesis

Interpret image brightness as a sound spectrum.



THEORY CONNECTION

Phase retrieval

Estimate the missing phase to synthesize audio.

$$X^{(r+1)} = A e^{j\angle \text{STFT}(x^{(r)})}$$

Synthesis is approximate.

Noise-reduction methods

Different ways to reduce unwanted sound.

Spectral subtraction

Subtract a noise estimate.

$$\hat{A} = \max(A - \alpha D, \beta A)$$

Spectral gate

Attenuate bins below a threshold.

$$\hat{X} = [(1 - \rho) + \rho M]X$$

Local Wiener

Use local mean and variance.

$$\hat{x} = \mu + \max(1 - \sigma_v^2 / \sigma_x^2, 0)(x - \mu)$$

Live spectrogram

Analyze incoming microphone audio.

THEORY CONNECTION

Windowed FFTs

Analyze short sections as new audio arrives.

1024

input frames per callback

8 S

rolling display history

150 ms

analysis timer interval

Inside a live effect preview

The callback plays buffers; the background worker computes DSP.

- 1 Control changes** Coalesce parameter updates on an 80 ms timer.
- 2 Worker renders** Process the committed source for the selected range.
- 3 Latest result wins** Ignore a finished job if its generation is stale.
- 4 Buffer changes** Keep the playback position and crossfade for 10 ms.
- 5 Apply commits** Store one undoable edit. Export writes a file.

Preview latency includes DSP time. This is buffer-based preview, not a hard real-time DSP engine.

Sampling and Nyquist

Sample rate limits the available frequencies.

THEORY CONNECTION

Sampling theorem

Sample above twice the highest signal frequency.

$$F_s > 2f_{\max}$$



Measured results

Deterministic fixtures make the numerical behavior reproducible.

Experiment	Configuration	Observed result
Gain	-6 dB	Amplitude ratio 0.501187
Butterworth low-pass	Order 4, fc 1000 Hz, Fs 16000 Hz	-42.10 dB at 3000 Hz
Peaking EQ	1000 Hz, +6 dB, Q = 1.4	+6.000 dB at center
FFT	4096 samples, Fs 16000 Hz	500 Hz peak, amplitude 0.400
STFT / inverse STFT	Hann, N 512, hop 128, seed 42	Max error 1.11×10^{-16}
Noise subtraction	Two tones + Gaussian noise	SNR 9.7 dB to 22.1 dB

Noise test: Fs 16000 Hz, 2 s, noise SD 0.055, independent 1 s profile, N 2048, hop 512, strength 1.5.
SNR compares each output against the known clean signal. This is not a speech-quality benchmark.

Current boundaries

The implementation has clear limits and a practical next-step list.

In-memory audio

Mono and stereo files are loaded into memory. Long selections take more time to process and preview.

Long FFT selections

Overlapping windows preserve the original frequency scale. Averaging summarizes the selection; use the spectrogram for timing.

Memory budgets

Spectral processing and mixing have 256 MiB buffer budgets. Recording stops at 64 MiB of stored samples.

Signal artifacts

Strong noise reduction or abrupt spectral masks can affect sound quality. Image synthesis estimates missing phase.

Hardware validation

Tests simulate audio streams. Rehearse speakers, microphone and projector on the actual lab setup.

Sources and implementation

Theory references support the concepts. Repository code defines the delivered behavior.

[1] SonicCraft source code

github.com/mehrab4226/SonicCraft

[2] SciPy: Butterworth design and SOS filtering

docs.scipy.org/doc/scipy/reference/generated/scipy.signal.butter.html

[3] W3C: Audio EQ Cookbook, Robert Bristow-Johnson formulae

[w3.org/TR/audio-eq-cookbook/](https://www.w3.org/TR/audio-eq-cookbook/)

[4] NumPy: real-input discrete Fourier transform

numpy.org/doc/stable/reference/generated/numpy.fft.rfft.html

[5] SciPy: Short-Time Fourier Transform tutorial

docs.scipy.org/doc/scipy/tutorial/signal.html

[6] Aalto University: Introduction to Speech Processing

speechprocessingbook.aalto.fi/enhancement/noise-attenuation/

[7] Griffin-Lim algorithm reference and original 1984 paper citation

librosa.org/doc/main/api/generated/librosa.griffinlim.html

Local DSP: effects.py, filters.py, transforms.py, noise_reduction.py. UI and playback: desktop/.